

Midterm Project: A Nice Title for your Midterm Project

Your names

September 3, 2026

1 Introduction

Write down a nice introduction for your project...

2 Region of Interest (ROI) and Case Study

Provide a succinct geographic description of the selected event...

3 Methodology

Explain your implementation step by step: The implementation follows a structured five-phase workflow (Lillesand, Kiefer, and Chipman [2015](#))...

3.1 Phase 1: Problem Definition & Data Acquisition

- **Dates:** Before (t_1) and After (t_2) selection.
- **Data Sources:** Optical (Sentinel-2 or Landsat 8/9) and/or optional SAR (Sentinel-1).

3.2 Phase 2: Pre-processing

To ensure images are radiometrically comparable, the following steps were applied:

- **Atmospheric Correction:** ...
- **Co-registration:** ...
- **Resampling:** (if needed) ...
- **Cloud Masking:** ...

3.3 Phase 3: Image Processing & Analysis

We utilized specific methods based on the event type. Table [1](#) show those techniques.

Table 1: Applied Change Detection Methodologies

Method	Technique
...	...
...	...

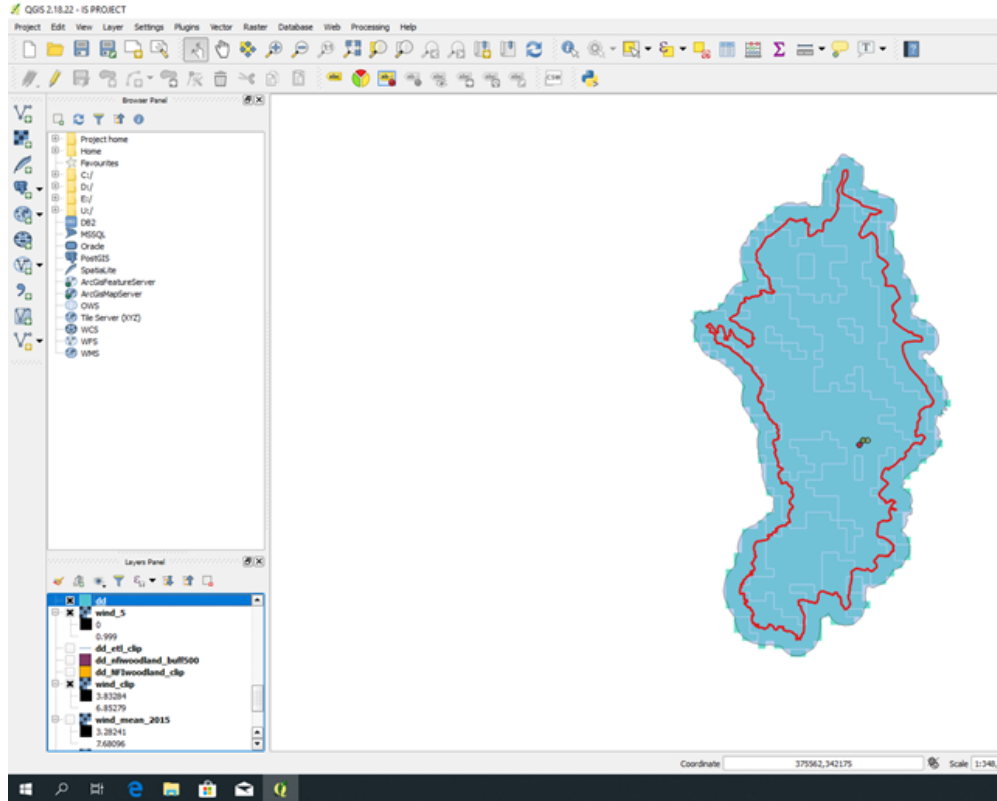


Figure 1: An example of vectorization in QGIS

3.4 Phase 4: Post-Processing & Accuracy Assessment

- **Thresholding:** ...
- **Validation:** ...
- **Vectorization:** See QGIS details in Figure 1 (QGIS Development Team 2026) ...

4 Results and Discussion

4.1 Thematic Maps

...

4.2 Statistics

...

4.3 Storytelling & Discussion

...

5 Conclusion

Write down a nice set of conclusions for this project.

References

- Lillesand, Thomas, Ralph W. Kiefer, and Jonathan Chipman (2015). *Remote Sensing and Image Interpretation*. 7th. Hoboken, NJ: John Wiley & Sons. ISBN: 978-1-118-34328-9.
- QGIS Development Team (2026). *QGIS Geographic Information System*. Open Source Geospatial Foundation. URL: <https://qgis.org>.